CERTIFICATE

Class: _____ Year: _____

This is to certify that Shri/Kumari _____, a student of _____, Division _____ Roll No. _____ Seat No. _____, has satisfactorily completed the required practical work recorded in this Journal. He/She has attended and completed the practical sessions for the 1st Term / 2nd Term / Both Terms during the academic year 20___ - 20___in the college laboratory as prescribed by the university. The work submitted has been found satisfactory.

_____	_____	_____
Head of the Department	External Examiner	Internal Examiner

Date: ___ / ___ / 20___

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Practical No. 1

Aim: Data Importing and Dataset Preparation

a. Import data from CSV and Excel files using read.csv() and readxl::read_excel().
b. Define and convert variable types (e.g., numeric, factor, character).
c. Explore dataset structure using str(), summary(), and glimpse().
d. Save and reload R datasets using write.csv(), saveRDS(), and readRDS().

Code:

```
# ==============================================
# Practical 1: Data Importing and Dataset Preparation in R
# ==============================================

# ---------------------------------------------
# Install Packages (Run only once)
# ---------------------------------------------
install.packages("readxl") install.packages("dplyr")

# ---------------------------------------------
# Load Libraries
# ---------------------------------------------
library(readxl) library(dplyr)

# ---------------------------------------------
# (a) Import Data from CSV and Excel Files
# ---------------------------------------------

# Step 1: Import CSV File
data_csv <- read.csv("students_practical.csv")

# Explanation:
# read.csv() reads comma-separated values
# Data is stored as a data frame
# File must be in working directory

# Check working directory getwd()

# Set working directory (if needed)
# setwd("C:/Users/YourName/Documents")

# Step 2: Import Excel File
data_excel <- read_excel("students_practical.xls")
```

```
# Explanation:
# read_excel() reads .xls or .xlsx files
# Automatically detects column names and data types
# Excel installation is not required


# ------------------------------------------- #
(b) Define and Convert Variable Types
# -------------------------------------------

# Step 3: View Current Data Types
str(data_csv)

# Explanation:
# Shows column names, data types, and sample values
# Important for understanding dataset

# Step 4: Convert Variable Types

# Character → Numeric
data_csv$Age <- as.numeric(data_csv$entry_age)

# Character → Factor
data_csv$Gender <- as.factor(data_csv$gender)

# Numeric → Character
data_csv$Student_ID <- as.character(data_csv$id)

# Explanation:
# as.numeric()   → for calculations
# as.factor()    → for categories
# as.character() → for text data
# Note: Wrong conversion may create NA values


# -------------------------------------------
# (c) Explore Dataset Structure
# -------------------------------------------

# Step 5: Check Structure Again
str(data_csv)
```

```
# Why this matters:
# Confirms conversions worked correctly

# Step 6: Summary Statistics summary(data_csv)

# Explanation:
# Numeric: Min, Max, Mean, Median
# Factor: Frequency of categories

# Step 7: Glimpse (Modern View) glimpse(data_csv)

# Explanation:
# Compact and readable format
# Best for large datasets


# ------------------------------------------
# (d) Save and Reload R Datasets
# ------------------------------------------

# Step 8: Save Dataset as CSV
write.csv(data_csv, "cleaned_students.csv", row.names = FALSE)

# Explanation:
# Saves as text file
# row.names = FALSE avoids extra index column

# Step 9: Save Dataset as RDS saveRDS(data_csv,
"students_data.rds")

# Explanation:
# R-specific format
# Faster and preserves data types

# Step 10: Reload RDS File new_data <-
readRDS("students_data.rds")

# Explanation:
# Restores dataset exactly


# ------------------------------------------
# Final Verification
```

# ---------- str(new_data)
summary(new_data)

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/
>
>
> # ----------------------------------------
> # Final Verification
> # ----------------------------------------
> str(new_data)
'data.frame':   10 obs. of  13 variables:
 $ id              : int  1 2 3 4 4 5 6 7 8 9
 $ first_name      : chr  "John" "Jane" "Ali" "Sarah" ...
 $ last_name       : chr  "Doe" "Smith" "Khan" "Thomas" ...
 $ gender          : chr  "M" "F" "M" "F" ...
 $ hs_gpa          : num  3.2 NA 2.8 3.6 3.6 2.1 NA 3.8 2.5 3
 $ entry_age       : num  18 18.5 17.8 NA NA 18.3 19 18.9 17.6 NA
 $ city            : chr  "Phoenix" "New York" "Phoenix" "Chicago" ...
 $ state           : chr  "Arizona" "New York" "Arizona" "Illinos" ...
 $ act_composite   : int  22 25 NA 21 21 18 24 26 NA 20
 $ first_generation: chr  "TRUE " "FALSE" "TRUE " "FALSE" ...
 $ Age             : num  18 18.5 17.8 NA NA 18.3 19 18.9 17.6 NA
 $ Gender          : Factor w/ 2 levels "F","M": 2 1 2 1 1 2 1 2 1 2
 $ Student_ID      : chr  "1" "2" "3" "4" ...
> summary(new_data)
       id        first_name         last_name           gender             hs_gpa
 Min.   :1.00   Length:10          Length:10          Length:10          Min.   :2.100
 1st Qu.:3.25   Class :character   Class :character   Class :character   1st Qu.:2.725
 Median :4.50   Mode  :character   Mode  :character   Mode  :character   Median :3.100
 Mean   :4.90                                                           Mean   :3.075
 3rd Qu.:6.75                                                           3rd Qu.:3.600
 Max.   :9.00                                                           Max.   :3.800
                                                                        NA's   :2
```

cleaned_students.csv

Sheet 1

cleaned_students

id	first_name	last_name	gender	hs_gpa	entry_age	city	state	act_composite	first_generation	Age	Gender	Student_ID
1	John	Doe	M	3.2	18	Phoenix	Arizona	22	TRUE	18	M	1
2	Jane	Smith	F	NA	18.5	New York	New York	25	FALSE	18.5	F	2
3	Ali	Khan	M	2.8	17.8	Phoenix	Arizona	NA	TRUE	17.8	M	3
4	Sarah	Thomas	F	3.6	NA	Chicago	Illinos	21	FALSE	NA	F	4
4	Sarah	Thomas	F	3.6	NA	Chicago	Illinos	21	FALSE	NA	F	4
5	Ahmed	Patel	M	2.1	18.3	Houston	Texas	18	TRUE	18.3	M	5
6	Emily	Brown	F	NA	19	Denver	Colorado	24	FALSE	19	F	6
7	David	Wilson	M	3.8	18.9	Seattle	Washington	26	TRUE	18.9	M	7
8	Ananya	Sharma	F	2.5	17.6	Mumbai	Maharastra	NA	FALSE	17.6	F	8
9	Michael	Lee	M	3	NA	San Jose	California	20	TRUE	NA	M	9

Practical No. 2

Aim: Data Cleaning and Missing Value Handling

a. Identify missing values using is.na() and visualize with visdat or naniar.
b. Impute missing values using tidyr::replace_na() or basic mean substitution.
c. Remove duplicates using dplyr::distinct().
d. Standardize variables using scale() and apply z-score transformations.

Code:

```
# ==============================================
# Practical: Data Cleaning and Preprocessing in R
# ==============================================
install.packages("tidyr", dependencies = TRUE) install.packages("visdat",
dependencies = TRUE)
install.packages("naniar", dependencies = TRUE)
# -------------------------------------------
# Load Required Libraries
# -------------------------------------------
library(dplyr) library(tidyr)
library(visdat) library(naniar)

# -------------------------------------------
# Set Working Directory
# -------------------------------------------
setwd('/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical 2')

# -------------------------------------------
# Load Dataset
# ------------------------------------------- data
<- read.csv("students_practical.csv")

# -------------------------------------------
# 1. Visualize Missing Data
# -------------------------------------------
vis_miss(data) gg_miss_var(data)

# Explanation:
# vis_miss()     → Shows missing values visually
# gg_miss_var()  → Shows missing values per column

# -------------------------------------------
# 2. Check Data Types
# -------------------------------------------
str(data[, c("hs_gpa", "entry_age")])
```

```
# Explanation:
# Displays structure of selected columns

# ------------------------------------------
# 3. Convert to Numeric
# ------------------------------------------
data$hs_gpa <- as.numeric(data$hs_gpa) data$entry_age
<- as.numeric(data$entry_age)

# Note:
# Invalid values may become NA

# ------------------------------------------
# 4. Impute Missing Values
# ------------------------------------------
data$hs_gpa[is.na(data$hs_gpa)] <- mean(data$hs_gpa, na.rm = TRUE)
data$entry_age[is.na(data$entry_age)] <- mean(data$entry_age, na.rm = TRUE)

# Explanation:
# Missing values replaced with mean of column

# ------------------------------------------
# 5. Remove Duplicate Records
# ------------------------------------------ data
<- distinct(data)

# Explanation:
# Removes duplicate rows

# ------------------------------------------
# 6. Standardize GPA
# ------------------------------------------
data$hs_gpa_scaled <- scale(data$hs_gpa)

# Explanation:
# Converts values to standard scale (mean = 0, sd = 1)

# ------------------------------------------
# 7. Final Verification
# ------------------------------------------
summary(data) str(data)
```

```
# ------------------------------------------ #
Save Cleaned Dataset
# ------------------------------------------
write.csv(   data,
  file = "/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
2/students_practical_cleaned.csv",
  row.names = FALSE
)
```

Output:

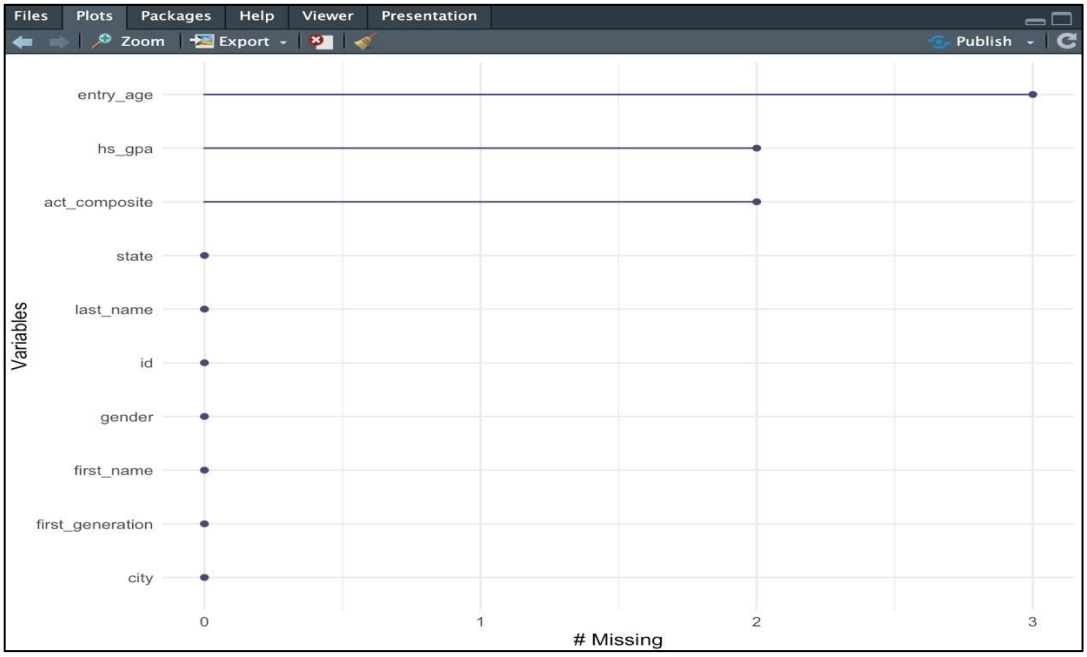
```
Console   Terminal ×   Background Jobs ×
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 2/
> str(data)
'data.frame':   9 obs. of  11 variables:
 $ id              : int  1 2 3 4 5 6 7 8 9
 $ first_name      : chr  "John" "Jane" "Ali" "Sarah" ...
 $ last_name       : chr  "Doe" "Smith" "Khan" "Thomas" ...
 $ gender          : chr  "M" "F" "M" "F" ...
 $ hs_gpa          : num  3.2 3.08 2.8 3.6 2.1 ...
 $ entry_age       : num  18 18.5 17.8 18.3 18.3 19 18.9 17.6 18.3
 $ city            : chr  "Phoenix" "New York" "Phoenix" "Chicago" ...
 $ state           : chr  "Arizona" "New York" "Arizona" "Illinos" ...
 $ act_composite   : int  22 25 NA 21 18 24 26 NA 20
 $ first_generation: chr  "TRUE " "FALSE" "TRUE " "FALSE" ...
 $ hs_gpa_scaled   : num [1:9, 1] 0.354 0.113 -0.418 1.126 -1.769 ...
  ..- attr(*, "scaled:center")= num 3.02
  ..- attr(*, "scaled:scale")= num 0.518
>
> # ------------------------------------------
> # Save Cleaned Dataset
> # ------------------------------------------
> write.csv(
+   data,
+   file = "/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical 2/students_practical_cleaned.cs
v",
+   row.names = FALSE
+ )
```

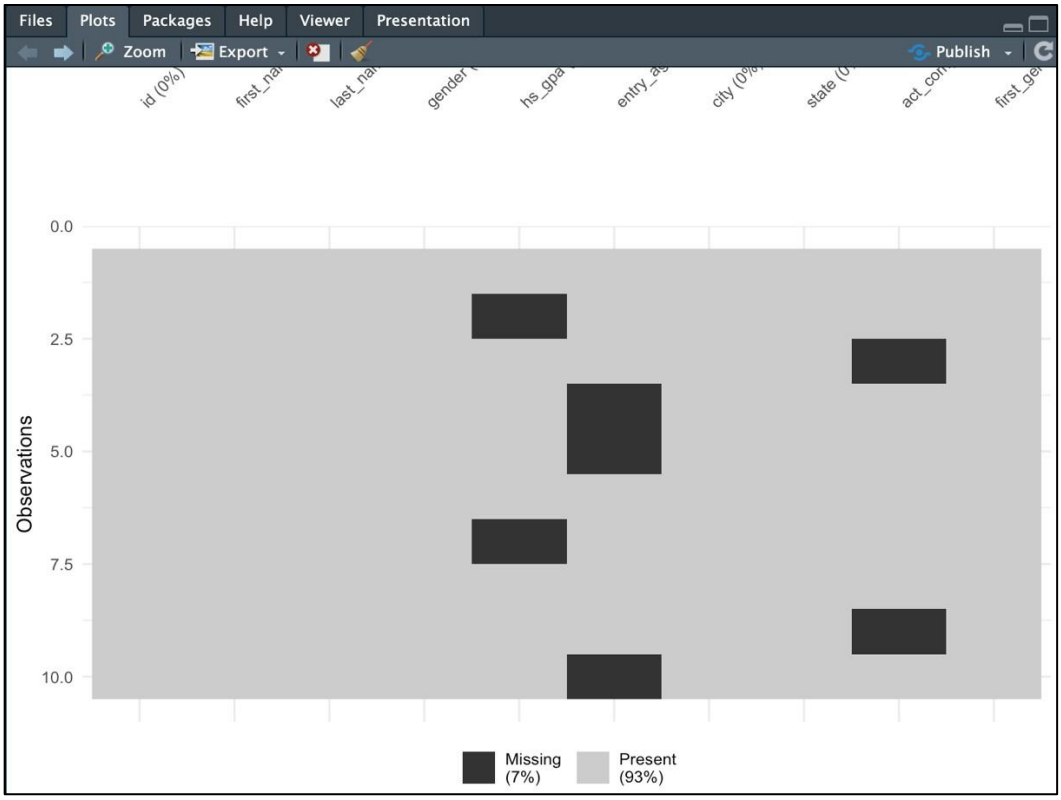
students_practical_cleaned.csv

Sheet 1

students_practical_cleaned

id	first_name	last_name	gender	hs_gpa	entry_age	city	state	act_composite	first_generation	hs_gpa_scaled
1	John	Doe	M	3.2	18	Phoenix	Arizona	22	TRUE	0.353747584809629
2	Jane	Smith	F	3.075	18.5	New York	New York	25	FALSE	0.112556049712155
3	Ali	Khan	M	2.8	17.8	Phoenix	Arizona	NA	TRUE	-0.418065327502288
4	Sarah	Thomas	F	3.6	18.3	Chicago	Illinos	21	FALSE	1.12556049712155
5	Ahmed	Patel	M	2.1	18.3	Houston	Texas	18	TRUE	-1.76873792404814
6	Emily	Brown	F	3.075	19	Denver	Colorado	24	FALSE	0.112556049712155
7	David	Wilson	M	3.8	18.9	Seattle	Washington	26	TRUE	1.5114669532775
8	Ananya	Sharma	F	2.5	17.6	Mumbai	Maharastra	NA	FALSE	-0.996925011736225
9	Michael	Lee	M	3	18.3	San Jose	California	20	TRUE	-0.0321588713463297




Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
id (0%)
first_na
last_na
gender
hs_gpa
entry_a
city (0%
state (0
act_com
first_ge
Observations
0.0
2.5
5.0
7.5
10.0
Missing (7%)
Present (93%)

Practical No. 3

Aim: Data Transformation and Recoding

a. Create new variables using mutate() and case_when().
b. Recode categorical variables using forcats::fct_recode() or dplyr::recode().
c. Merge datasets using left_join() and concatenate with bind_rows().
d. Reshape data using pivot_longer() and pivot_wider() from tidyr.

Code:

```
# working directory
setwd("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical 3")

# -------------------------------------------
# LOAD DATA
# -------------------------------------------

data <- read.csv(
  "/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical 3/students_practical.csv"
)

# -------------------------------------------
# LOAD LIBRARIES
# -------------------------------------------

library(dplyr)
library(tidyr)   # Needed for pivot_longer and pivot_wider

# -------------------------------------------
# CREATE AGE GROUP
# -------------------------------------------

data <- data %>%
mutate(
    age_group = case_when(
entry_age < 18 ~ "Below 18",
      entry_age >= 18 & entry_age <= 19 ~ "18-19",
entry_age > 19 ~ "Above 19"
    )
  )
```

```
# ------------------------------------------
# CREATE PERFORMANCE CATEGORY
# ------------------------------------------

data <- data %>%
mutate(
   performance = case_when(
     hs_gpa >= 3.5 ~ "Excellent",       hs_gpa
>= 2.5 & hs_gpa < 3.5 ~ "Good",
     hs_gpa < 2.5 ~ "Average"
   )
  )

# ------------------------------------------
# RECODE GENDER
# ------------------------------------------

data <- data %>%
mutate(     gender
= recode(
gender,
     "M" = "Male",
     "F" = "Female"
   )
  )

# ------------------------------------------
# ADD SCHOLARSHIP DATA
# ------------------------------------------

scholarship <- data.frame(
  id = c(1, 3, 5, 7),
  scholarship = c("Yes", "No", "Yes", "Yes")
)

data <- left_join(data, scholarship, by = "id")

# ------------------------------------------
# ADD NEW STUDENTS
# ------------------------------------------

new_students <- data.frame(   id = c(10,
11),   first_name = c("Ravi", "Neha"),
```

```
last_name = c("Patil", "Joshi"),   gender =
c("Male", "Female"),   hs_gpa = c(3.1,
2.9),   entry_age = c(18.2, 18.6),   city =
c("Nagpur", "Nashik"),   state =
c("Maharashtra", "Maharashtra"),
act_composite = c(21, 23),
  first_generation = c("Yes", "No")
)

data <- bind_rows(data, new_students)

# -------------------------------------------
# CONVERT WIDE TO LONG FORMAT
# -------------------------------------------

long_data <- data %>%
pivot_longer(     cols =
c(hs_gpa, entry_age),
names_to = "variable",
values_to = "value"
  )

long_data

# -------------------------------------------
# CONVERT LONG TO WIDE FORMAT
# -------------------------------------------

wide_data <- long_data %>%
pivot_wider(     names_from
= variable,     values_from =
value
  )
wide_data
```

Console Terminal Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 3/

```
> long_data
# A tibble: 24 × 13
      id first_name last_name gender city     state    act_composite first_generation
   <dbl> <chr>      <chr>     <chr>  <chr>    <chr>            <dbl> <chr>
 1     1 John       Doe       Male   Phoenix  Arizona             22 "TRUE "
 2     1 John       Doe       Male   Phoenix  Arizona             22 "TRUE "
 3     2 Jane       Smith     Female New York New York            25 "FALSE"
 4     2 Jane       Smith     Female New York New York            25 "FALSE"
 5     3 Ali        Khan      Male   Phoenix  Arizona             NA "TRUE "
 6     3 Ali        Khan      Male   Phoenix  Arizona             NA "TRUE "
 7     4 Sarah      Thomas    Female Chicago  Illinos             21 "FALSE"
 8     4 Sarah      Thomas    Female Chicago  Illinos             21 "FALSE"
 9     4 Sarah      Thomas    Female Chicago  Illinos             21 "FALSE"
10     4 Sarah      Thomas    Female Chicago  Illinos             21 "FALSE"
```

Console Terminal Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 3/

```
> wide_data
# A tibble: 11 × 13
      id first_name last_name gender city     state      act_composite first_generation
   <dbl> <chr>      <chr>     <chr>  <chr>    <chr>              <dbl> <chr>
 1     1 "John"     Doe       Male   Phoenix  "Arizona"             22 "TRUE "
 2     2 "Jane"     Smith     Female New York "New York"            25 "FALSE"
 3     3 "Ali"      Khan      Male   Phoenix  "Arizona"             NA "TRUE "
 4     4 "Sarah"    Thomas    Female Chicago  "Illinos"             21 "FALSE"
 5     5 "Ahmed"    Patel     Male   Houston  "Texas"               18 "TRUE "
 6     6 "Emily"    Brown     Female Denver   "Colorado"            24 "FALSE"
 7     7 "David"    Wilson    Male   Seattle  "Washingt…            26 "TRUE "
 8     8 "Ananya "  Sharma    Female Mumbai   "Maharast…            NA "FALSE"
 9     9 "Michael"  Lee       Male   San Jose "Californ…            20 "TRUE "
10    10 "Ravi"     Patil     Male   Nagpur   "Maharash…            21 "Yes"
11    11 "Neha"     Joshi     Female Nashik   "Maharash…            23 "No"
```

Sheet 1

students_practical

id	first_name	last_name	gender	hs_gpa	entry_age	city	state	act_composite	first_generation
1	John	Doe	M	3.2	18	Phoenix	Arizona	22	TRUE
2	Jane	Smith	F		18.5	New York	New York	25	FALSE
3	Ali	Khan	M	2.8	17.8	Phoenix	Arizona		TRUE
4	Sarah	Thomas	F	3.6		Chicago	Illinos	21	FALSE
4	Sarah	Thomas	F	3.6		Chicago	Illinos	21	FALSE
5	Ahmed	Patel	M	2.1	18.3	Houston	Texas	18	TRUE
6	Emily	Brown	F		19	Denver	Colorado	24	FALSE
7	David	Wilson	M	3.8	18.9	Seattle	Washington	26	TRUE
8	Ananya	Sharma	F	2.5	17.6	Mumbai	Maharastra		FALSE
9	Michael	Lee	M	3		San Jose	California	20	TRUE

Practical No. 4

Aim: Filtering, Subsetting, and Conditional Logic

a. Filter rows with conditions using filter() (e.g., Age > 30 & Gender == "Male").
b. Sort datasets using arrange().
c. Select and drop columns using select() and select(-colname).
d. Apply grouped operations using group_by() and summarise().

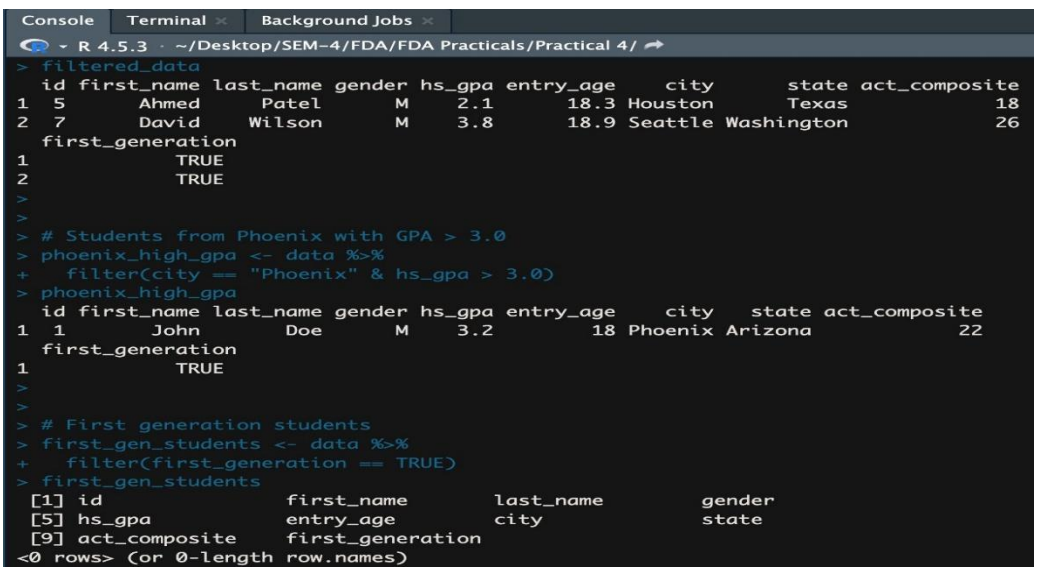
Code:

```
# -------------------------------------------
# INSTALL & LOAD LIBRARY
# -------------------------------------------

install.packages("dplyr")   # Run only first time library(dplyr)

# -------------------------------------------
# LOAD DATA
# -------------------------------------------

data <- read.csv("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
4/students_practical.csv")

# -------------------------------------------
# FILTER DATA
# ------------------------------------------- #
Students older than 18 and Male
filtered_data <- data %>%
filter(entry_age > 18 & gender == "M")
filtered_data

# Students from Phoenix with GPA > 3.0
phoenix_high_gpa <- data %>%
filter(city == "Phoenix" & hs_gpa > 3.0)
phoenix_high_gpa

# First generation students first_gen_students
<- data %>%   filter(first_generation ==
TRUE) first_gen_students
```

```
# ------------------------------------------
# SORTING DATA
# ------------------------------------------

# Ascending GPA sorted_gpa_asc
<- data %>%   arrange(hs_gpa)
sorted_gpa_asc

# Descending GPA sorted_gpa_desc
<- data %>%
arrange(desc(hs_gpa))
sorted_gpa_desc

# Sort by city and age sorted_city_age
<- data %>%   arrange(city,
entry_age)
sorted_city_age
# ------------------------------------------
# SELECT / DROP COLUMNS
# ------------------------------------------ #
Select specific columns selected_cols
<- data %>%
  select(first_name, last_name, gender, hs_gpa, entry_age) selected_cols

# Drop one column dropped_act
<- data %>%   select(-
act_composite)
dropped_act

# Drop multiple columns drop_multiple
<- data %>%
  select(-city, -state) drop_multiple
# ------------------------------------------
# GROUP BY & SUMMARISE
# ------------------------------------------

# Average GPA by gender avg_gpa_by_gender
<- data %>%   group_by(gender) %>%
  summarise(avg_hs_gpa = mean(hs_gpa, na.rm = TRUE)) avg_gpa_by_gender

# Count students by city count_by_city
<- data %>%   group_by(city) %>%
summarise(total_students = n())
count_by_city
```

```
# Average ACT score by first generation avg_act_by_firstgen
<- data %>%   group_by(first_generation) %>%
  summarise(avg_act = mean(act_composite, na.rm = TRUE)) avg_act_by_firstgen

# Overall summary summary_data
<- data %>%
  summarise(     avg_gpa = mean(hs_gpa,
na.rm = TRUE),     avg_age = mean(entry_age,
na.rm = TRUE),     count = n()
  )
summary_data
# -------------------------------------------
# EXPORT DATA
# ------------------------------------------- write.csv(
  avg_gpa_by_gender,
  "/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/avg_gpa_by_gender.csv",
row.names = FALSE
)
```

Output:



```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/
> filtered_data
  id first_name last_name gender hs_gpa entry_age    city      state act_composite
1  5      Ahmed     Patel      M    2.1      18.3 Houston      Texas            18
2  7      David    Wilson      M    3.8      18.9 Seattle Washington            26
  first_generation
1             TRUE
2             TRUE
>
>
> # Students from Phoenix with GPA > 3.0
> phoenix_high_gpa <- data %>%
+   filter(city == "Phoenix" & hs_gpa > 3.0)
> phoenix_high_gpa
  id first_name last_name gender hs_gpa entry_age    city   state act_composite
1  1       John       Doe      M    3.2         18 Phoenix Arizona            22
  first_generation
1             TRUE
>
>
> # First generation students
> first_gen_students <- data %>%
+   filter(first_generation == TRUE)
> first_gen_students
 [1] id               first_name       last_name        gender
 [5] hs_gpa           entry_age        city             state
 [9] act_composite    first_generation
<0 rows> (or 0-length row.names)
```

Console Terminal Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/

```
> sorted_gpa_asc
   id first_name last_name gender hs_gpa entry_age     city       state act_composite
1   5      Ahmed     Patel      M    2.1      18.3  Houston       Texas            18
2   8     Ananya    Sharma      F    2.5      17.6   Mumbai  Maharastra            NA
3   3        Ali      Khan      M    2.8      17.8  Phoenix     Arizona            NA
4   9    Michael       Lee      M    3.0        NA San Jose California            20
5   1       John       Doe      M    3.2      18.0  Phoenix     Arizona            22
6   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
7   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
8   7      David    Wilson      M    3.8      18.9  Seattle  Washington            26
9   2       Jane     Smith      F     NA      18.5 New York    New York            25
10  6      Emily     Brown      F     NA      19.0   Denver    Colorado            24
   first_generation
1              TRUE
2             FALSE
3              TRUE
4              TRUE
5              TRUE
6             FALSE
7             FALSE
8              TRUE
9             FALSE
10            FALSE
```

Console Terminal Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/

```
> # Descending GPA
> sorted_gpa_desc <- data %>%
+   arrange(desc(hs_gpa))
> sorted_gpa_desc
   id first_name last_name gender hs_gpa entry_age     city       state act_composite
1   7      David    Wilson      M    3.8      18.9  Seattle  Washington            26
2   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
3   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
4   1       John       Doe      M    3.2      18.0  Phoenix     Arizona            22
5   9    Michael       Lee      M    3.0        NA San Jose California            20
6   3        Ali      Khan      M    2.8      17.8  Phoenix     Arizona            NA
7   8     Ananya    Sharma      F    2.5      17.6   Mumbai  Maharastra            NA
8   5      Ahmed     Patel      M    2.1      18.3  Houston       Texas            18
9   2       Jane     Smith      F     NA      18.5 New York    New York            25
10  6      Emily     Brown      F     NA      19.0   Denver    Colorado            24
   first_generation
1              TRUE
2             FALSE
3             FALSE
4              TRUE
5              TRUE
6              TRUE
7             FALSE
8              TRUE
9             FALSE
10            FALSE
```

```
Console   Terminal ×   Background Jobs ×
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/
> sorted_city_age
   id first_name last_name gender hs_gpa entry_age     city       state act_composite
1   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
2   4      Sarah    Thomas      F    3.6        NA  Chicago     Illinos            21
3   6      Emily     Brown      F     NA      19.0   Denver    Colorado            24
4   5      Ahmed     Patel      M    2.1      18.3  Houston       Texas            18
5   8    Ananya     Sharma      F    2.5      17.6   Mumbai  Maharastra            NA
6   2       Jane     Smith      F     NA      18.5 New York    New York            25
7   3        Ali      Khan      M    2.8      17.8  Phoenix     Arizona            NA
8   1       John       Doe      M    3.2      18.0  Phoenix     Arizona            22
9   9    Michael       Lee      M    3.0        NA San Jose California            20
10  7      David    Wilson      M    3.8      18.9  Seattle  Washington            26
   first_generation
1             FALSE
2             FALSE
3             FALSE
4              TRUE
5             FALSE
6             FALSE
7              TRUE
8              TRUE
9              TRUE
10             TRUE
>
>
> # ------------------------------------------
> # SELECT / DROP COLUMNS
> # ------------------------------------------
>
> # Select specific columns
> selected_cols <- data %>%
+   select(first_name, last_name, gender, hs_gpa, entry_age)
> selected_cols
   first_name last_name gender hs_gpa entry_age
1        John       Doe      M    3.2      18.0
2        Jane     Smith      F     NA      18.5
3         Ali      Khan      M    2.8      17.8
4       Sarah    Thomas      F    3.6        NA
5       Sarah    Thomas      F    3.6        NA
6       Ahmed     Patel      M    2.1      18.3
7       Emily     Brown      F     NA      19.0
8       David    Wilson      M    3.8      18.9
9     Ananya     Sharma      F    2.5      17.6
10    Michael       Lee      M    3.0        NA
```

Sheet 1

avg_gpa_by_gender

gender	avg_hs_gpa
F	3.23333333333333
M	2.98

Console | Terminal | Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 4/

```
> count_by_city
# A tibble: 8 × 2
  city     total_students
  <chr>             <int>
1 Chicago               2
2 Denver                1
3 Houston               1
4 Mumbai                1
5 New York              1
6 Phoenix               2
7 San Jose              1
8 Seattle               1
>
>
> # Average ACT score by first generation
> avg_act_by_firstgen <- data %>%
+   group_by(first_generation) %>%
+   summarise(avg_act = mean(act_composite, na.rm = TRUE))
> avg_act_by_firstgen
# A tibble: 2 × 2
  first_generation avg_act
  <chr>              <dbl>
1 "FALSE"             22.8
2 "TRUE "             21.5
>
>
> # Overall summary
> summary_data <- data %>%
+   summarise(
+     avg_gpa = mean(hs_gpa, na.rm = TRUE),
+     avg_age = mean(entry_age, na.rm = TRUE),
+     count = n()
+   )
> summary_data
  avg_gpa avg_age count
1   3.075    18.3    10
```

Practical No. 5

Aim: Descriptive Statistics and Distribution Analysis

a. Calculate mean, median, mode, variance, and standard deviation using base R and psych package.
b. Generate frequency tables using table() and janitor::tabyl().
c. Visualize data distributions with hist(), boxplot(), and ggplot2.
d. Interpret summary statistics with summary() and skimr::skim().

Code:

```
# ==============================================
# PRACTICAL 5: DESCRIPTIVE STATISTICS & DISTRIBUTION ANALYSIS
# ==============================================

# AIM:
# To calculate descriptive statistics and analyze data distribution using R.

# ---------------------------------------------
# STEP 1: LOAD LIBRARIES (Run install once)
# ---------------------------------------------
install.packages("psych") install.packages("janitor")
install.packages("ggplot2")
install.packages("skimr")

library(psych) library(janitor)
library(ggplot2)
library(skimr)

# ---------------------------------------------
# STEP 2: LOAD DATA
# ---------------------------------------------
data <- read.csv("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
5/students_practical.csv")

# ---------------------------------------------
# (a) DESCRIPTIVE STATISTICS
# ---------------------------------------------

# Mean
mean(data$hs_gpa, na.rm = TRUE)

# Median
median(data$hs_gpa, na.rm = TRUE)
```

```
# Mode (custom function) mode_func
<- function(x) {
  ux <- unique(x)
  ux[which.max(tabulate(match(x, ux)))]
}
mode_func(data$hs_gpa)

# Variance
var(data$hs_gpa, na.rm = TRUE)

# Standard Deviation
sd(data$hs_gpa, na.rm = TRUE)

# Using psych package
describe(data$hs_gpa)
```

```
Console   Terminal ×   Background Jobs ×
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 5/
> # (a) DESCRIPTIVE STATISTICS
> # ----------------------------------------
>
> # Mean
> mean(data$hs_gpa, na.rm = TRUE)
[1] 3.075
>
> # Median
> median(data$hs_gpa, na.rm = TRUE)
[1] 3.1
>
> # Mode (custom function)
> mode_func <- function(x) {
+   ux <- unique(x)
+   ux[which.max(tabulate(match(x, ux)))]
+ }
> mode_func(data$hs_gpa)
[1] NA
>
> # Variance
> var(data$hs_gpa, na.rm = TRUE)
[1] 0.3507143
>
> # Standard Deviation
> sd(data$hs_gpa, na.rm = TRUE)
[1] 0.5922114
>
> # Using psych package
> describe(data$hs_gpa)
   vars n mean   sd median trimmed  mad min max range  skew kurtosis   se
X1    1 8 3.08 0.59    3.1    3.08 0.74 2.1 3.8   1.7 -0.28    -1.53 0.21
```

```
# ----------------------------------------
# (b) FREQUENCY TABLES
# ----------------------------------------
```

# Using table()
table(data$gender)

Using janitor tabyl(data$gender)

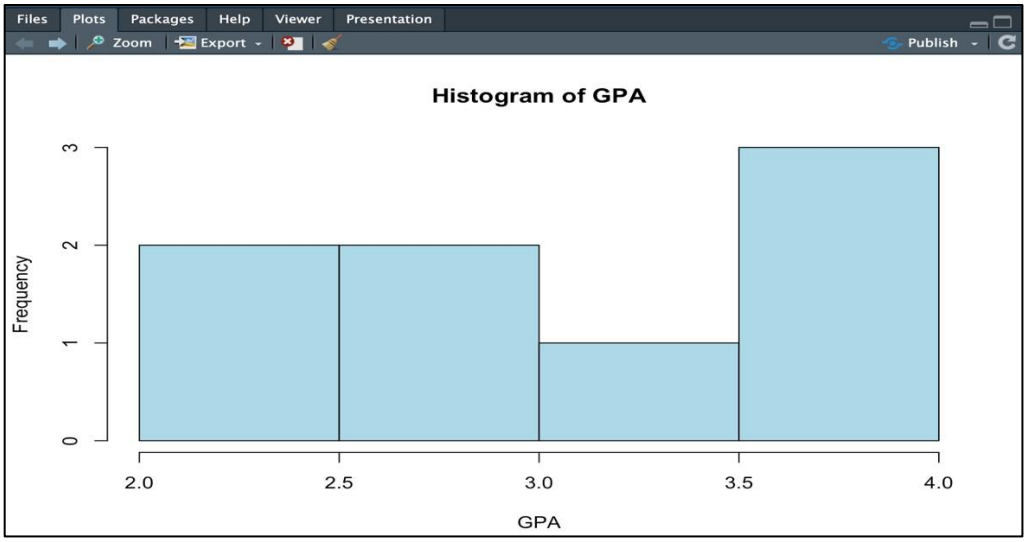
```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 5/
> # (b) FREQUENCY TABLES
> # ----------------------------------------
>
> # Using table()
> table(data$gender)

F M
5 5
>
> # Using janitor
> tabyl(data$gender)
 data$gender n percent
           F 5     0.5
           M 5     0.5
```

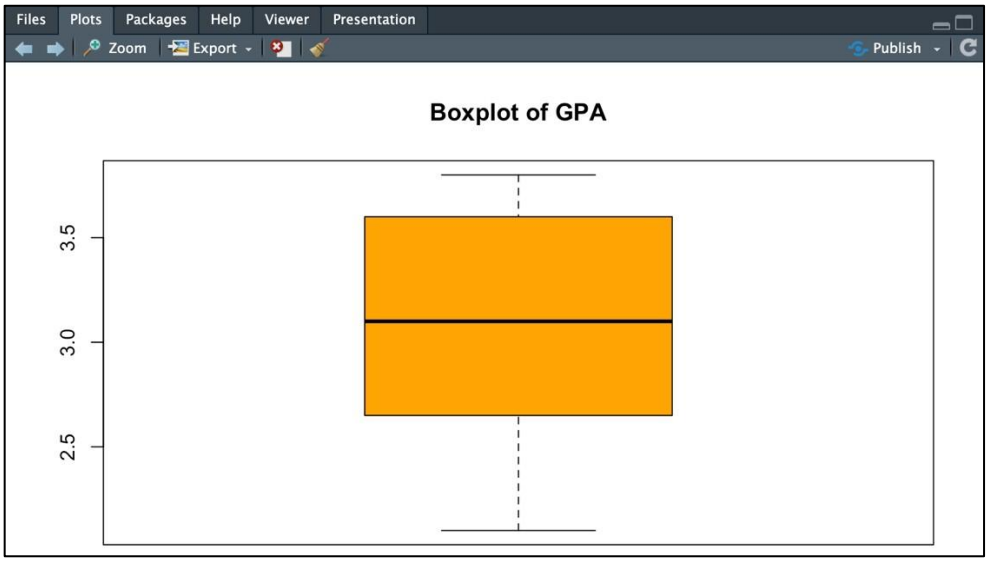
(c) DATA VISUALIZATION

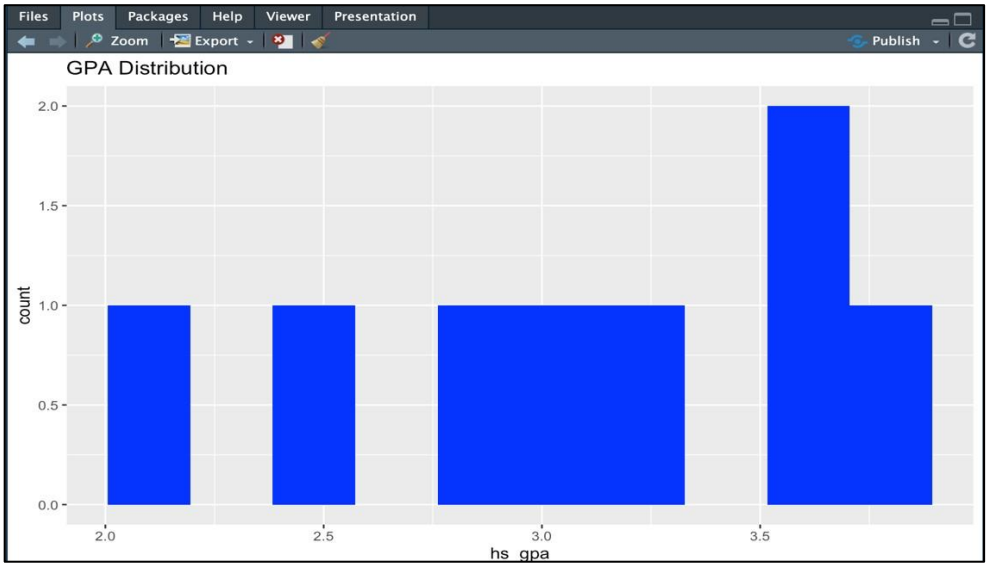
Histogram (Base R)
hist(
data$hs_gpa,
  main = "Histogram of GPA",
  col = "lightblue",
xlab = "GPA"
)

Boxplot boxplot(
data$hs_gpa, main =
"Boxplot of GPA",
  col = "orange"
)

ggplot Histogram ggplot(data, aes(x = hs_gpa)) + geom_histogram(fill = "blue", bins = 10) + ggtitle("GPA Distribution")


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Histogram of GPA
Frequency
0
1
2
3
2.0
2.5
3.0
3.5
4.0
GPA


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Boxplot of GPA
2.5
3.0
3.5


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
GPA Distribution
count
0.0
0.5
1.0
1.5
2.0
2.0
2.5
3.0
3.5
hs_gpa

```
# ------------------------------------------
# (d) DATA INTERPRETATION
# ------------------------------------------
# Summary statistics
summary(data$hs_gpa)

# Detailed summary using skimr skim(data)

# ------------------------------------------
# CONCLUSION
# ------------------------------------------
# Descriptive statistics such as mean, median, #
variance, and standard deviation were calculated. #
Frequency tables and visualizations helped #
understand data distribution.
# The dataset was summarized successfully.
```

```
Console   Terminal ×   Background Jobs ×
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 5/
> # (d) DATA INTERPRETATION
> # ------------------------------------------
>
> # Summary statistics
> summary(data$hs_gpa)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  2.100   2.725   3.100   3.075   3.600   3.800       2
>
> # Detailed summary using skimr
> skim(data)
── Data Summary ────────────────────────
                           Values
Name                       data
Number of rows             10
Number of columns          10
_______________________
Column type frequency:
  character                6
  numeric                  4
_______________________
Group variables            None

── Variable type: character ────────────────────────────────────────────────────────────
  skim_variable    n_missing complete_rate min max empty n_unique whitespace
1 first_name               0            1   3   7     0        9          0
2 last_name                0            1   3   6     0        9          0
3 gender                   0            1   1   1     0        2          0
4 city                     0            1   6   8     0        8          0
5 state                    0            1   5  11     0        8          0
6 first_generation         0            1   5   5     0        2          0

── Variable type: numeric ──────────────────────────────────────────────────────────────
  skim_variable n_missing complete_rate  mean    sd   p0   p25  p50   p75 p100 hist
1 id                    0           1    4.9  2.60   1    3.25  4.5  6.75  9   ▅▅▅▅▅▅▅▅
2 hs_gpa                2           0.8  3.08 0.592  2.1  2.72  3.1  3.6   3.8 ▂▅▃▅▅▅▅▅
3 entry_age             3           0.7 18.3  0.535 17.6 17.9  18.3 18.7  19  ▅▅▅▅▅▅▅▅
4 act_composite         2           0.8 22.1  2.70  18   20.8  21.5 24.2  26  ▅▅▅▅▅▅▅▅
```

Practical No. 6

Aim: Hypothesis Testing – Mean and Median a.
Perform a one-sample t-test using t.test().
b. Conduct an independent samples t-test using t.test(x ~ group).
c. Use paired sample t-test for pre/post analysis.
d. Apply non-parametric test like wilcox.test() (Mann-Whitney U test).

Code:

```
# ==========================================
# PRACTICAL 6: HYPOTHESIS TESTING – MEAN AND MEDIAN
# ==========================================

# AIM:
# To perform hypothesis testing using t-tests and non-parametric tests in R.

# ------------------------------------------
# STEP 1: LOAD DATA
# ------------------------------------------

data <- read.csv('/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
6/students_practical.csv')
# Loads dataset into R as a dataframe named "data"
# ------------------------------------------
# (a) ONE-SAMPLE T-TEST
# ------------------------------------------

t.test(data$hs_gpa, mu = 3.0)
# Tests whether mean GPA is equal to 3.0
# hs_gpa = sample data
# mu = hypothesized population mean
```

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 6/
> t.test(data$hs_gpa, mu = 3.0)

        One Sample t-test

data:  data$hs_gpa
t = 0.3582, df = 7, p-value = 0.7307
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 2.579899 3.570101
sample estimates:
mean of x
    3.075
```

 # (b) INDEPENDENT SAMPLE T-TEST

----

t.test(hs_gpa ~ gender, data = data)
# Compares mean GPA between two groups (Male vs Female)
# hs_gpa = numeric variable
# gender = grouping variable

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 6/
> t.test(hs_gpa ~ gender, data = data)

        Welch Two Sample t-test

data:  hs_gpa by gender
t = 0.55171, df = 4.2351, p-value = 0.609
alternative hypothesis: true difference in means between group F and group M is not equal to
0
95 percent confidence interval:
 -0.9941177  1.5007843
sample estimates:
mean in group F mean in group M
       3.233333        2.980000
```

 # (c) PAIRED SAMPLE T-TEST

Example: Pre-test and Post-test scores
pre <- c(60, 65, 70, 75, 80) post <-
c(65, 70, 75, 78, 85)
t.test(pre, post, paired = TRUE) #
Compares before and after values
paired = TRUE means same subjects are measured twice

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 6/
> t.test(pre, post, paired = TRUE)

        Paired t-test

data:  pre and post
t = -11.5, df = 4, p-value = 0.0003264
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -5.710578 -3.489422
sample estimates:
mean difference
           -4.6
```

 # (d) NON-PARAMETRIC TEST (WILCOX TEST)

----

wilcox.test(hs_gpa ~ gender, data = data)
# Performs Mann-Whitney U test
# Used when data is NOT normally distributed
# Compares median between groups

OUTPUT INTERPRETATION (REFERENCE)
----
p-value:
If p-value < 0.05 → Reject null hypothesis # If
p-value ≥ 0.05 → Fail to reject null hypothesis #
confidence interval:
Range in which true mean lies
t value:
Test statistic used for decision making

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 6/
> wilcox.test(hs_gpa ~ gender, data = data)

        Wilcoxon rank sum test with continuity correction

data:  hs_gpa by gender
W = 9, p-value = 0.7642
alternative hypothesis: true location shift is not equal to 0
```

```
# ------------------------------------------
# CONCLUSION:
# ------------------------------------------
# One-sample, independent, and paired t-tests #
were performed to compare means.
# Wilcoxon test was used for median comparison #
when assumptions are not satisfied.
```

Practical No. 7

Aim: ANOVA and MANOVA

a. Conduct one-way ANOVA using aov() and interpret results.
b. Run two-way ANOVA with interaction terms.
c. Perform MANOVA using manova() and interpret Wilks' Lambda.
d. Apply Tukey's HSD post-hoc test with TukeyHSD().

Code:

```
# ==========================================
# PRACTICAL 7: ANOVA AND MANOVA
# ==========================================

# AIM:
# To perform one-way ANOVA, two-way ANOVA, MANOVA,
# and post-hoc analysis in R.
# ------------------------------------------
# STEP 1: LOAD DATA
# ------------------------------------------

data <- read.csv("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
7/students_practical.csv")
# Loads dataset into R
# ------------------------------------------
# (a) ONE-WAY ANOVA
# ------------------------------------------

anova_model <- aov(hs_gpa ~ gender, data = data)
# hs_gpa = dependent variable
# gender = independent variable

summary(anova_model)
# Displays ANOVA table (F-value, p-value)

# ------------------------------------------
# INTERPRETATION
# ------------------------------------------
# If p-value < 0.05 → Reject H0 → Means are different
# If p-value ≥ 0.05 → Fail to reject H0 → Means are same
```

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 7/
> summary(anova_model)
            Df Sum Sq Mean Sq F value Pr(>F)
gender       1 0.1203  0.1203   0.309  0.598
Residuals    6 2.3347  0.3891
2 observations deleted due to missingness
```

 # (b) TWO-WAY ANOVA

----

anova2_model <- aov(hs_gpa ~ gender * first_generation, data = data)

gender = factor 1

first_generation = factor 2

* includes interaction effect

summary(anova2_model)

Shows main effects and interaction effect

INTERPRETATION

Check p-values for:

1. gender effect

2. first_generation effect

3. interaction effect

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 7/
> summary(anova2_model)
            Df Sum Sq Mean Sq F value Pr(>F)
gender       1 0.1203  0.1203   0.309  0.598
Residuals    6 2.3347  0.3891
2 observations deleted due to missingness
```

 # (c) MANOVA

----

manova_model <- manova(

cbind(hs_gpa, act_composite) ~ gender,

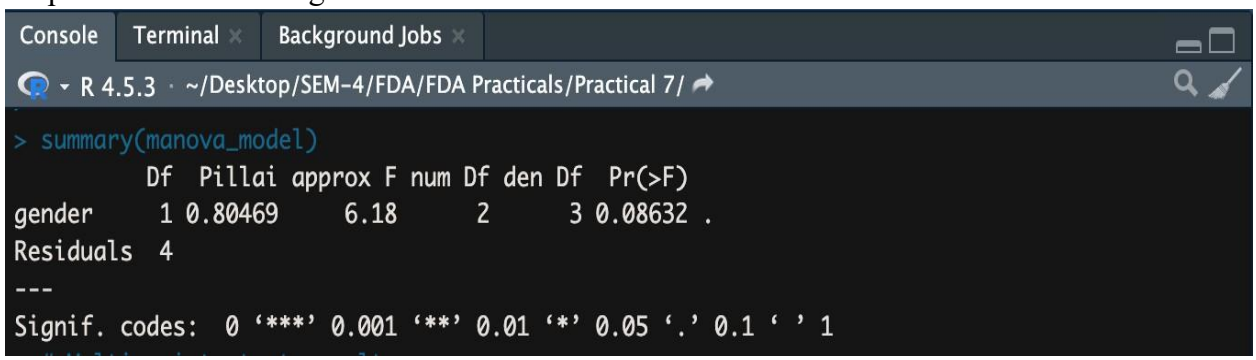
data = data

)

Multiple dependent variables:

hs_gpa and act_composite

summary(manova_model)
# Multivariate test results
# ----------
# WILKS' LAMBDA INTERPRETATION
# ---------- #
Small value → Groups are different
Large value → Groups are similar
If p-value < 0.05 → Significant effect



 # (d) TUKEY POST-HOC TEST

TukeyHSD(anova_model)
# Pairwise comparison between groups

INTERPRETATION

If p-value < 0.05 → Significant difference
Shows which groups differ

CONCLUSION

ANOVA compares group means
MANOVA analyzes multiple dependent variables
Tukey test identifies specific group differences

Console Terminal Background Jobs

R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 7/

```
> TukeyHSD(anova_model)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = hs_gpa ~ gender, data = data)

$gender
          diff       lwr       upr     p adj
M-F -0.2533333 -1.368024 0.8613576 0.5982502
```

Practical No. 8

Aim: Chi-Square and Categorical Data Analysis

a. Generate contingency tables using table() and xtabs().
b. Perform Chi-square test using chisq.test() and interpret p-values.
c. Calculate expected counts and contribution to Chi-square value.
d. Visualize results using ggplot2::geom_bar() and stacked bar plots.

Code:

```
# ==========================================
# PRACTICAL 9: CHI-SQUARE AND CATEGORICAL DATA ANALYSIS
# ==========================================

# AIM:
# To perform Chi-square test and analyze categorical data using R

# ------------------------------------------
# STEP 1: LOAD DATA
# ------------------------------------------
data <- read.csv("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
9/students_practical.csv")
# Loads dataset into R as a dataframe

# ------------------------------------------
# (a) CONTINGENCY TABLES
# ------------------------------------------

table(data$gender, data$first_generation) #
Creates contingency table (frequency count)

xtabs(~ gender + first_generation, data = data)
# Alternative formula-based method
```

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 9/
> # (a) CONTINGENCY TABLES
> # ----------------------------------------
>
> table(data$gender, data$first_generation)

    FALSE TRUE
  F     5    0
  M     0    5
> # Creates contingency table (frequency count)
>
> xtabs(~ gender + first_generation, data = data)
      first_generation
gender FALSE TRUE
     F     5    0
     M     0    5
```

 # (b) CHI-SQUARE TEST

----

chisq.test(table(data$gender, data$first_generation)) #
Tests whether gender and first_generation are related

INTERPRETATION

If p-value < 0.05 → Reject H0 → Variables are dependent

If p-value ≥ 0.05 → Fail to reject H0 → Variables are independent

```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 9/
> # (b) CHI-SQUARE TEST
> # ----------------------------------------
>
> chisq.test(table(data$gender, data$first_generation))

        Pearson's Chi-squared test with Yates' continuity correction

data:  table(data$gender, data$first_generation)
X-squared = 6.4, df = 1, p-value = 0.01141
```

 # (c) EXPECTED COUNTS & CONTRIBUTION

----

chisq_result <- chisq.test(

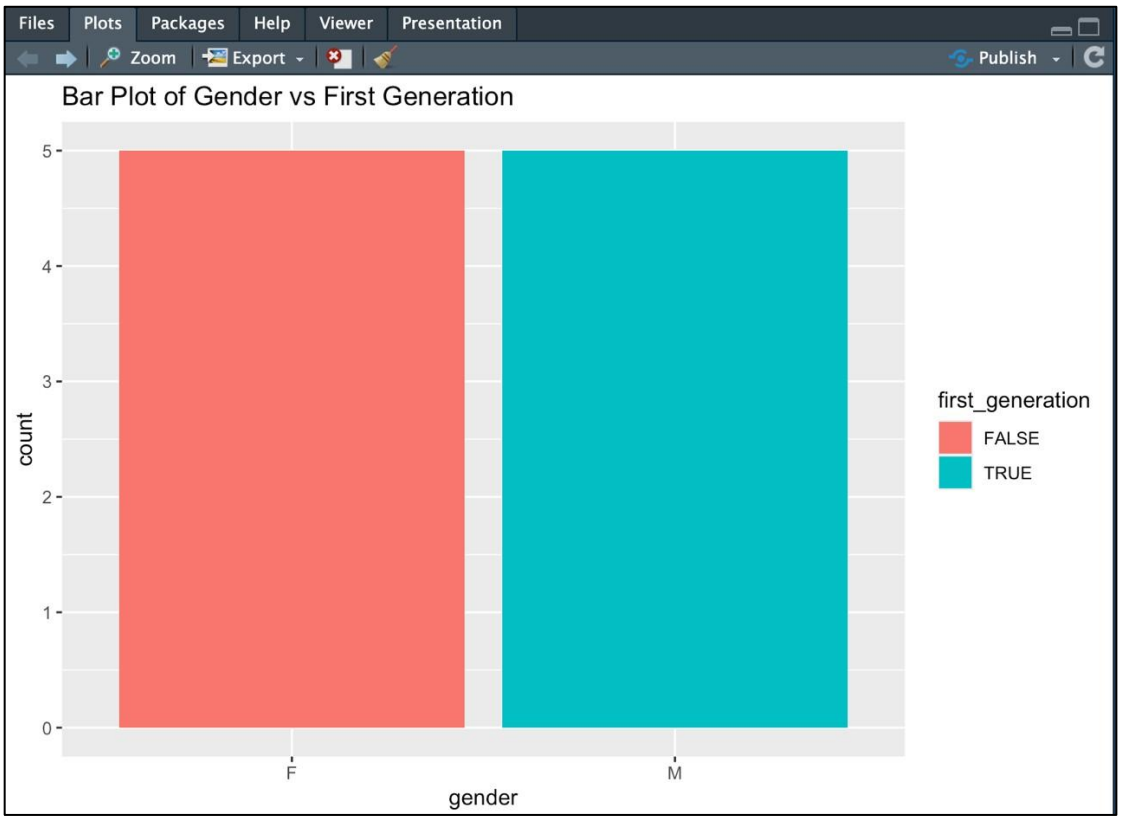
```
  table(data$gender, data$first_generation)
)

chisq_result$expected
# Expected frequencies for each cell

chisq_result$residuals
# Contribution of each cell to Chi-square value
# Larger values = higher contribution
```

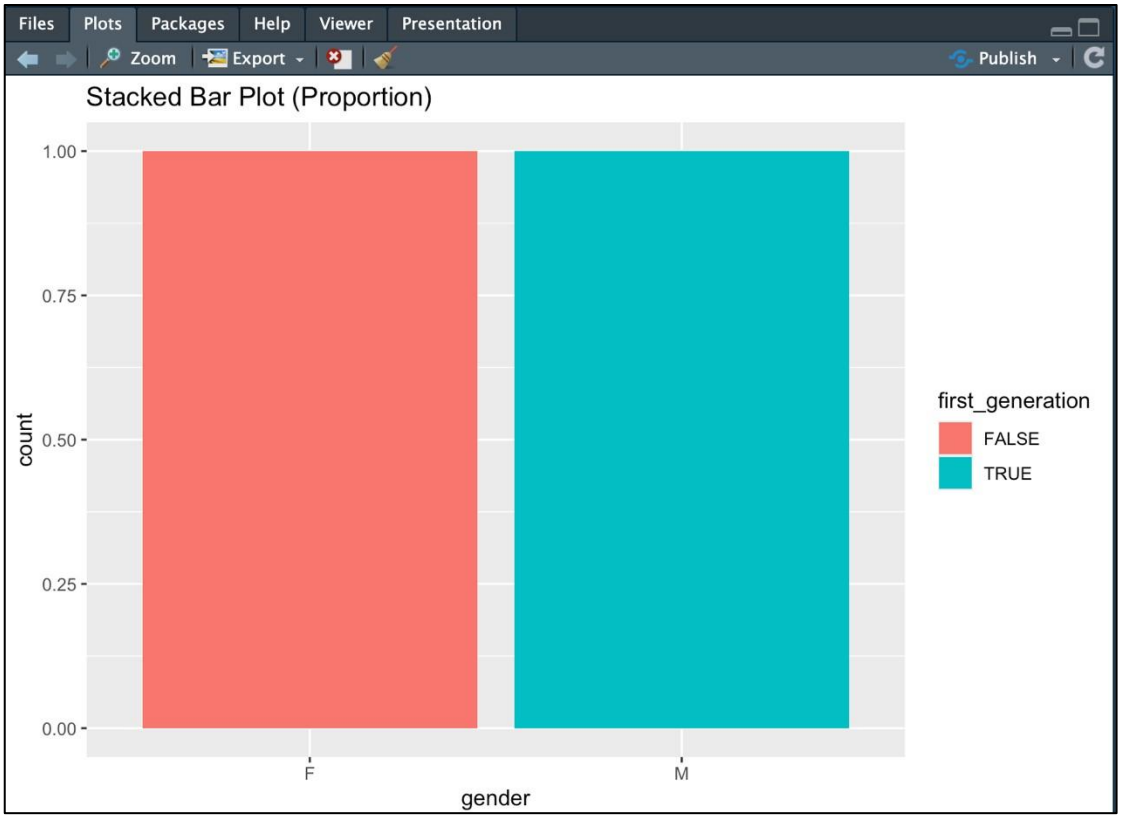
```
Console   Terminal   Background Jobs
R 4.5.3 · ~/Desktop/SEM-4/FDA/FDA Practicals/Practical 9/
> # (c) EXPECTED COUNTS & CONTRIBUTION
> # ----------------------------------------
>
> chisq_result <- chisq.test(
+   table(data$gender, data$first_generation)
+ )

Warning message:
In chisq.test(table(data$gender, data$first_generation)) :
  Chi-squared approximation may be incorrect

>
> chisq_result$expected

    FALSE TRUE
  F   2.5   2.5
  M   2.5   2.5
> # Expected frequencies for each cell
>
> chisq_result$residuals

        FALSE     TRUE
  F  1.581139 -1.581139
  M -1.581139  1.581139
```

```
# ----------------------------------------
# (d) DATA VISUALIZATION
# ----------------------------------------

library(ggplot2)

# Grouped bar chart
ggplot(data, aes(x = gender, fill = first_generation)) +
geom_bar(position = "dodge") +
  ggtitle("Bar Plot of Gender vs First Generation")

# Stacked bar chart (proportion)
ggplot(data, aes(x = gender, fill = first_generation)) +
geom_bar(position = "fill") +   ggtitle("Stacked Bar
Plot (Proportion)")
```

CONCLUSION

Chi-square test analyzes relationship between categorical variables
Expected counts and residuals explain contribution
Bar plots help visualize categorical distribution





Practical No. 9

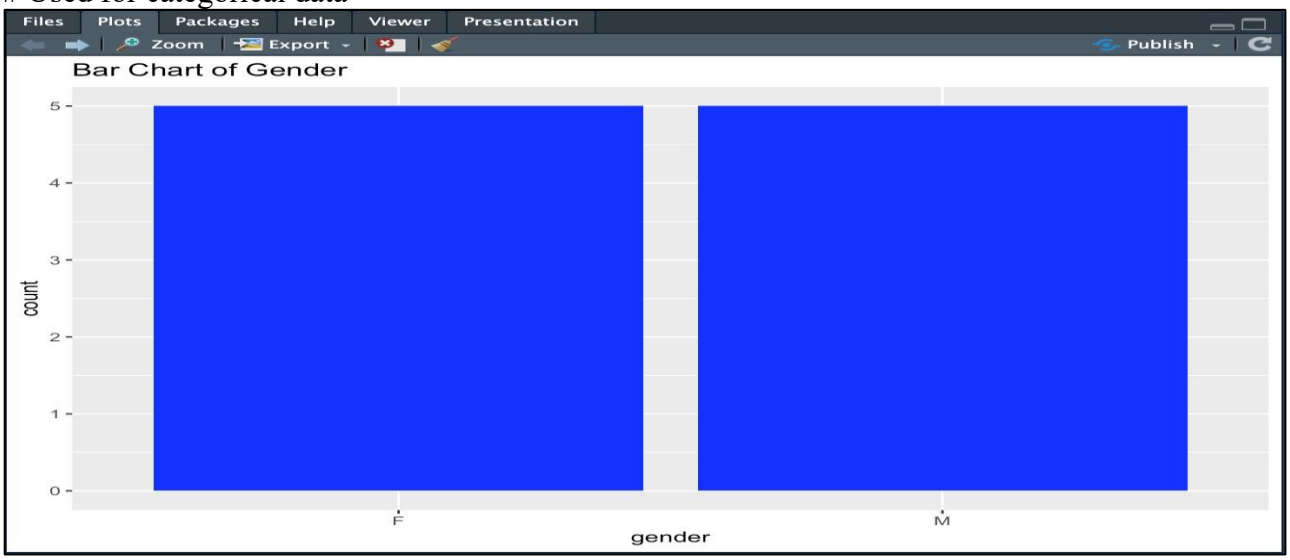
Aim: Visualization and Reporting

a. Create bar charts, line graphs, and histograms using ggplot2.

Code:

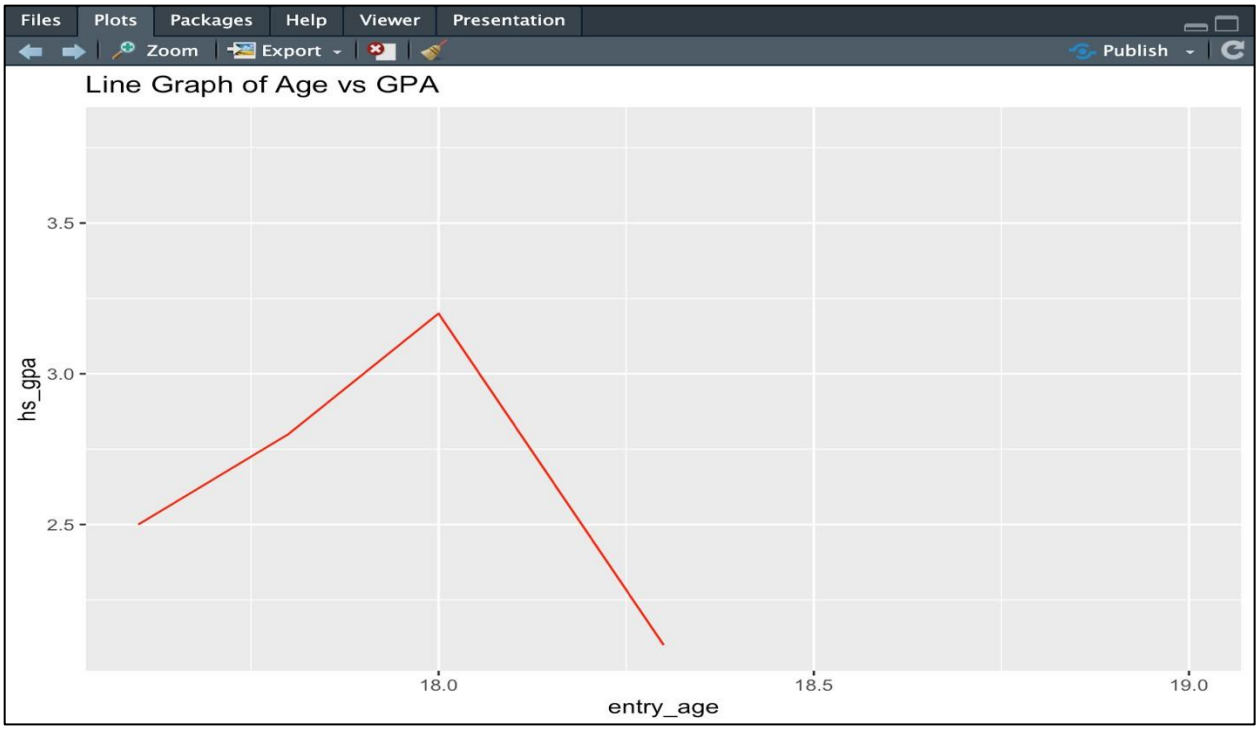
```
# ==============================================
# PRACTICAL 10: VISUALIZATION AND REPORTING #
============================================== #
AIM:
# To create bar charts, line graphs, and histograms using ggplot2 in R
# ------------------------------------------
# STEP 1: LOAD LIBRARY
# ------------------------------------------ install.packages("ggplot2")
# Run only first time library(ggplot2)          # Load package
# ------------------------------------------
# STEP 2: LOAD DATA
# ------------------------------------------
data <- read.csv("/Users/amitsharma/Desktop/SEM-4/FDA/FDA Practicals/Practical
10/students_practical.csv")
# Loads dataset into R
# ------------------------------------------
# (a) BAR CHART
```

```
# ------------------------------------------
ggplot(data, aes(x = gender)) +
geom_bar(fill = "blue") +
  ggtitle("Bar Chart of Gender")

# Shows count of males and females
# Used for categorical data
```

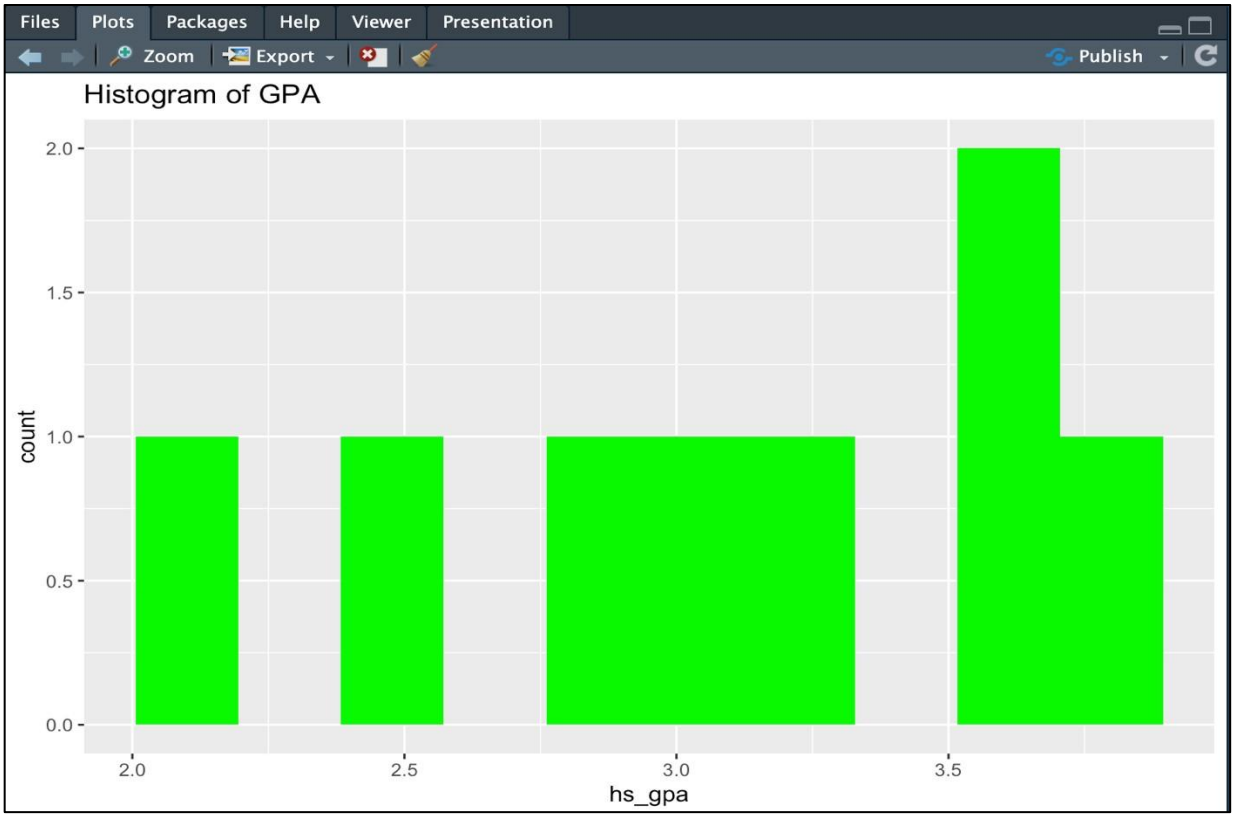



```
# ------------------------------------------
# (b) LINE GRAPH
# ------------------------------------------

ggplot(data, aes(x = entry_age, y = hs_gpa)) +
geom_line(color = "red") +
  ggtitle("Line Graph of Age vs GPA")

# Shows trend between age and GPA
```

Useful for ordered data




```
# ------------------------------------------
# (c) HISTOGRAM
# ------------------------------------------

ggplot(data, aes(x = hs_gpa)) +
geom_histogram(fill = "green", bins = 10) +
  ggtitle("Histogram of GPA")

# Shows distribution of GPA
# Helps identify spread and skewness
```




```
# ------------------------------------------
# CONCLUSION
# ------------------------------------------

# Bar charts → categorical data
# Line graphs → trends/relationships
# Histograms → data distribution
```